

group	số dòng	đã điền val_macro_f1	val_macro_f1 cao nhất	val_macro_f1 thấp nhất	val_acc cao nhất	đã thử chủ đề này?	nhận xét ngắn (bạn viết)
baseline	3	3	0,8569	0,8520	0,9101	—	Best val F1: base-s2 = 0.856881. GPU-memory=N/A on CPU.
loss	1	1	0,7361	0,7361	0,8711	Có	Best val F1: loss-mse = 0.736066. GPU-memory=N/A on CPU. Prediction: MSE on raw logits versus one-hot targets may fit classification less effectively than CE. Raw-logit MSE vs one-hot; ln(7) step0-gap formula is only meaningful for CE.
optimizer	3	3	0,8740	0,8264	0,9160	Có	Best val F1: opt-adam-0.003 = 0.874035. GPU-memory=N/A on CPU. Prediction: Larger Adam LR may accelerate early fitting but increase oscillation.
hparam	1	1	0,8095	0,8095	0,8822	Có	Best val F1: batch-2048 = 0.809498. GPU-memory=N/A on CPU. Prediction: Larger batch reduces updates per epoch and may slow convergence at fixed LR.
dropout	1	1	0,8418	0,8418	0,8982	Có	Best val F1: drop-0.1 = 0.841766. GPU-memory=N/A on CPU. Prediction: Dropout may reduce the train-val gap but hurt a baseline still improving at epoch 20.
clipping	3	3	0,8278	0,7441	0,8983	Có	Best val F1: clip-normal = 0.827755. GPU-memory=N/A on CPU. Prediction: Clipping should activate at normal LR and may slow learning. Clipped 14538/14540 steps.
amp	1	1	0,8500	0,8500	0,9071	Có	Best val F1: amp-bf16 = 0.850041. CPU GPU-memory=N/A; unavailable: FP16 training requires CUDA; measured BF16 on CPU. Prediction: Mixed precision may not improve end-to-end speed on this small MLP; numerical trajectories may change. Wall-clock mean includes long execution interruption; median=65.670s. See REPORT.
init	3	3	0,8493	0,0936	0,9037	Có	Best val F1: init-normal = 0.849264. GPU-memory=N/A on CPU. Prediction: Small normal weights shrink activations and may delay convergence.
final	3	3	0,8985	0,8887	0,9306	—	Best val F1: final-s3 = 0.898530. GPU-memory=N/A on CPU. Selected from opt-adam-0.003 by validation; extended 40 epochs.
other	0	0				—	
lr_screen	3	3	0,7062	0,5986	0,8460	—	Best val F1: lr-0.1-s1 = 0.706247. GPU-memory=N/A on CPU.

Số chủ đề (trong 7) đã thử: 7
7 chủ đề: loss, optimizer, hparam, dropout, clipping, amp, init. Một chủ đề 'đã thử' khi có ≥ 1 dòng có val_macro_f1.